

Highlights

Surface roughness build-up as an indicator of room-temperature creep

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- Interrupted 525 and 575 MPa tests resolve roughness–strain coupling.
- Roughening is front-loaded and becomes more sublinear at higher stress.
- Uninterrupted 500 and 530 MPa tests show spatial coarsening during creep.
- Simulated slip hotspots form early and persist during the creep hold.
- The solid-boundary model misses experimental short-wavelength PSD content.

Surface roughness build-up as an indicator of room-temperature creep

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ABSTRACT

Room-temperature creep accumulates over hundreds of hours, complicating monitoring. We test topography as a non-destructive creep indicator in laser-powder-bed-fused 316L stainless steel. Interrupted and uninterrupted tests combine digital image correlation and optical profilometry with EBSD-informed large-strain elasto-viscoplastic FFT simulations. Height maps are evaluated by arithmetic mean roughness and a 1 μm -sampled, $256 \times 128 \mu\text{m}^2$ observation operator. Because solid-only boundaries are not free surfaces, simulations are compared using normalized amplitude, spatial rank, power spectral density (PSD), and autocorrelation (ACF).

Interrupted 525 and 575 MPa tests resolve roughness–strain slopes of 0.346 and 0.257 μm per percent strain, whereas the 475 MPa interval includes zero. Roughness accumulation is sublinear in strain, with power-law exponents of 0.748 and 0.477 at 525 and 575 MPa: surface relief develops disproportionately early, especially at the higher stress. Uninterrupted 500 and 530 MPa tests also show spatial coarsening: spectral median wavelength rises by 26.3 and 34.5 μm and ACF length by 6.44 and 8.82 μm . The solid-boundary simulation reproduces the experimental correlation-length scale but allocates too little normalized power to short wavelengths. At 575 MPa, the simulated high-slip pattern is established by 5 h and then amplified, with an early-to-final spatial rank correlation of 0.978. Orthogonal EBSD sections constrain the reconstructed three-dimensional grain size, shape, and orientation through forward sectioning. The experimental roughness band, 4.13–60.87 μm , sets a $128 \times 128 \times 256 \mu\text{m}^3$ production domain with 1 μm voxels. The combined evidence supports baseline-referenced amplitude and broadband spatial redistribution as complementary indicators of creep state while identifying missing short-wavelength content in the periodic solid model.

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1. Introduction

Creep is the time-dependent accumulation of inelastic deformation under sustained stress. Although commonly associated with elevated temperature, measurable creep can occur near room temperature when the applied stress approaches the material's yield strength [1, 2, 3]. At low homologous temperature, dislocation climb is suppressed and time-dependent glide past comparatively weak obstacles can control the response [4, 2]. The resulting strain may accumulate over hundreds of hours, making conventional test matrices slow and expensive, while even modest deformation can relax preload, alter alignment, and reduce structural margins [5, 6].

Laser-powder-bed-fused (LPBF) 316L stainless steel is a particularly relevant material because its layerwise thermal history produces multiscale morphological and crystallographic heterogeneity. Grain shape, texture, subgrain structure, and residual stress can all influence localization and mechanical anisotropy [7, 8]. Statistical reconstruction of experimentally measured morphology therefore provides a practical route to representative-volume crystal-plasticity calculations without claiming a one-to-one reconstruction of a particular surface [9]. For room-temperature creep, this microstructure sensitivity creates both a challenge and an opportunity: surface relief may vary among nominally identical gauge regions, yet that same relief may record the heterogeneous deformation state.

Direct characterization of creep mechanisms is often destructive or spatially sparse. Slip traces, lattice rotation, and dislocation structures can reveal the active mechanisms, but they are poorly suited to repeated monitoring of large specimen populations. High-throughput geometries address the throughput problem by applying a common force to multiple gauge regions or stress levels [10, 11, 12, 13, 3]. Optical profilometry adds a complementary measurement: it is non-contact, repeatable, and sensitive to the cumulative surface displacement created by heterogeneous subsurface slip. The central unresolved question is whether topography contains a reproducible creep-state signal after accounting for initial surface condition, measurement scale, and microstructural variability.

Microstructure-resolved FFT methods provide a natural numerical counterpart. The original elasto-viscoplastic FFT formulation efficiently solves periodic polycrystal boundary-value problems on voxel grids [14]; its large-strain extension tracks finite deformation and evolving microstructure while retaining the computational advantages of FFT-based compatibility enforcement [15, 16]. Such calculations produce both macroscopic strain and spatially resolved slip, plastic strain, and boundary displacement. A periodic solid-only boundary plane is not a traction-free surface, however, so agreement in scalar roughness amplitude cannot be assumed. Mechanically meaningful comparisons must distinguish absolute experimental roughness from normalized progression, spatial organization, and within-model correlations.

This work combines those complementary levels of evidence. First, repeated interrupted tests quantify the absolute relationship between roughness accumulation and strain and resolve the time evolution of PSD and ACF, using the specimen rather than local map sections as the statistical unit. Second, less artifact-prone uninterrupted tests determine whether the association and spatial changes survive removal of unload/reload cycles. Third, orthogonal EBSD sections constrain a statistically reconstructed three-dimensional microstructure through forward sectioning, and microstructural sampling determines the calibration-domain dimensions. The resulting hierarchy

supports surface roughness as a baseline-referenced creep indicator while establishing an experimentally constrained basis for microstructure-resolved simulation.

2. Methods

2.1. Experimental setup

LPBF 316L stainless steel was produced in a Renishaw AM400 and tested without post-build heat treatment. The high-throughput geometry placed 25 nominally independent gauge regions in series under a common force (Fig. 1). Both gauge faces were ground through 1200-grit SiC paper. One face was prepared for digital image correlation (DIC); the opposite face was retained for optical profilometry.

Interrupted polished campaigns were conducted at 475, 525, and 575 MPa, with unloading and surface measurement after cumulative exposures of 0.25, 1.25, 5.25, 21.25, 85.25, and 341.25 h. Uninterrupted polished campaigns were held for approximately 350 h at 500, 530, and 588 MPa. DIC images were acquired with 12.3-MP monochrome FLIR Grasshopper 3 cameras and processed with VIC-2D. The first image after loading was the reference, so the reported strain excludes elastic loading and inelastic strain accumulated during the loading ramp. Height maps were measured in the unloaded state with a Keyence VK-X3000 laser-scanning microscope. The native lateral spacings were $1.380\ \mu\text{m}$ at $10\times$ and $0.278\ \mu\text{m}$ at $50\times$; both exports contained 768×1024 height samples.

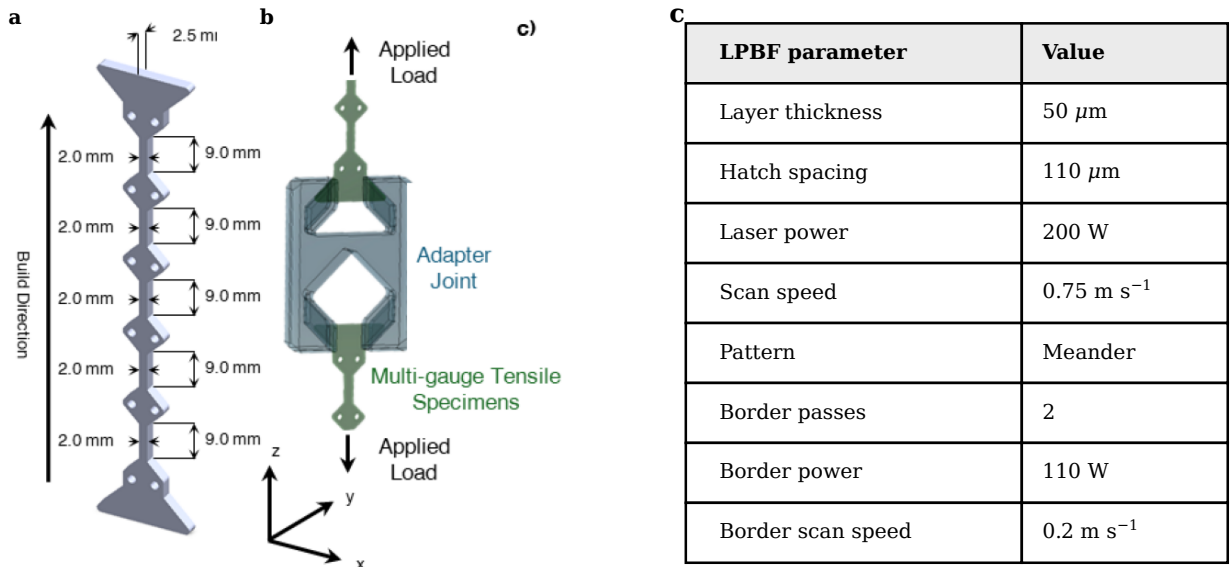


Figure 1: Experimental configuration. (a) Multi-gauge specimen geometry. (b) Serial loading of multi-gauge specimens. The specimen z axis is the LPBF build direction. (c) LPBF processing parameters.

2.2. DIC processing

An available-case mean formed independently at every DIC frame changes when a high- or low-straining specimen leaves the observable cohort. Interrupted histories also contain additive offsets when the DIC reference is reset. The mechanical target was therefore reconstructed from

within-specimen increments. For specimens finite at both ends of interval i ,

$$\Delta \bar{\epsilon}_i = \frac{1}{|S_i|} \sum_{s \in S_i} (\epsilon_{s,i} - \epsilon_{s,i-1}). \quad (1)$$

Cross-reset increments following each interrupted measurement were set to zero. The accumulated history was fitted on 256 points uniform in $\log(1 + t/1 \text{ min})$ by

$$\hat{\epsilon}(t) = \sum_{k=1}^{24} a_k [1 - \exp(-t/\tau_k)] + bt, \quad a_k \geq 0, \quad b \geq 0, \quad (2)$$

with fixed time constants spanning 20 s to three experiment durations. The nonnegative representation is continuous and monotone, preserves the paired-increment endpoint, and cannot manufacture a terminal acceleration. Two thousand whole-specimen bootstrap samples propagated between-specimen variation and the observed missingness pattern.

Figure 2 shows that the correction is not a generic smoothing of the data. Within each load, differences between the total mean and paired-increment mean isolate changing cohort composition and DIC resets; differences between the paired-increment mean and the target are the constrained interpolation alone. The largest relative correction occurs at 475 MPa because a comparatively high-straining specimen terminates at 20.75 h. Its missing tail is not filled: the resulting uncertainty remains in the whole-specimen bootstrap band. The detailed dropout and censoring audits are provided in Appendix A.

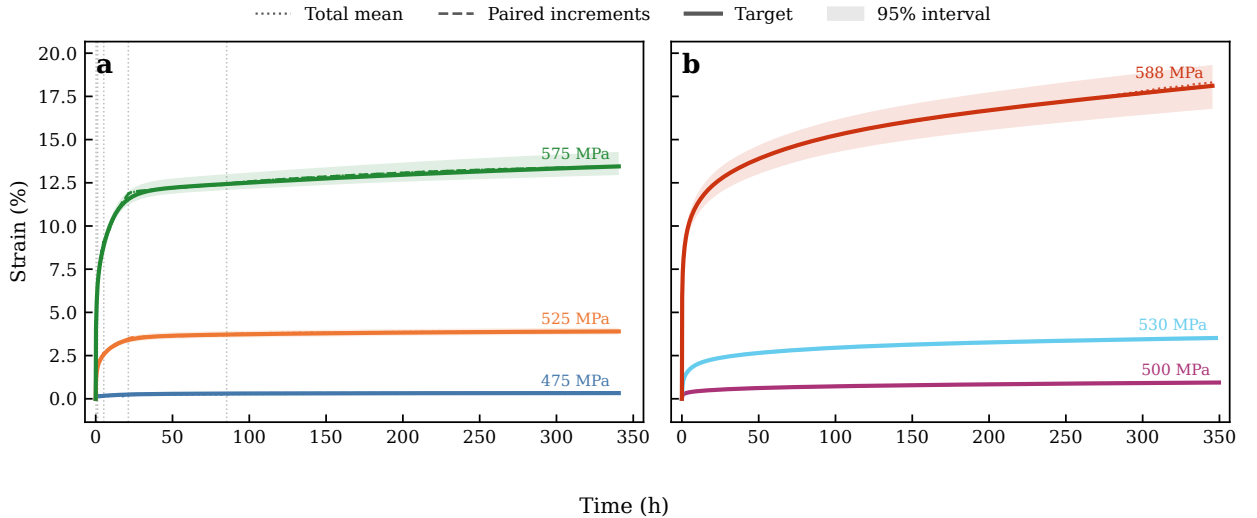


Figure 2: Construction of the six strain targets. (a) Interrupted 475, 525, and 575 MPa experiments. (b) Uninterrupted 500, 530, and 588 MPa experiments. Color identifies load in both panels. The total mean averages every specimen available at each time. The paired-increment mean averages only within-specimen changes between consecutive observations and cumulatively sums them, so specimen entry or dropout cannot create a step. The target is the constrained continuous interpolation of that history; shading is the whole-specimen 95% bootstrap interval. Vertical dotted lines mark interrupted-test DIC resets.

2.3. Roughness processing and retained length scales

Every profilometry export was required to have the declared calibration, exactly 768×1024 samples, a missing fraction no greater than 10^{-4} , and no connected missing region larger than one pixel. All 444 maps passed; 31 maps contained one isolated missing pixel, which was filled from the nearest finite neighbor. No map or local field was removed because its roughness, PSD, ACF, or height range was large.

A single plane over the full 1.06×1.41 mm $10\times$ field retains specimen-scale form and imperfectly centered fiducial relief: its median S_a is 1.453 times the selected local estimate. Each map was therefore divided into a 2×4 grid of eight $530 \times 353 \mu\text{m}^2$ fields and independently plane levelled. This is the only tested grid that provides at least eight subsamples, spans ten conservative correlation lengths on its short side, and contains at least 50 circular correlation areas. The specimen-time statistic is the median of the eight values

$$S_a = \frac{1}{N} \sum_{j=1}^N |h_j - \bar{h}|. \quad (3)$$

Sections are subsamples; the specimen remains the independent replicate.

Spatial morphology was evaluated on independently levelled $256 \times 128 \mu\text{m}^2$ windows. After mean removal and a separable Hann taper w , the area-scaled two-dimensional periodogram was

$$P(k_x, k_y) = \Delta x \Delta y \frac{|\mathcal{F}\{w(h - \bar{h})\}|^2}{\sum w^2}. \quad (4)$$

Radial PSDs used physical-frequency annuli. The ACF used zero-padded linear convolution with overlap-count correction, rather than circular correlation; its $1/e$ crossing describes the typical correlation length and its first zero crossing provides an independent upper-scale check.

The comparison band was selected from reproducible creep-associated change, not from microscope pixel size (Fig. 3). Let $C_f(\lambda)$ denote the radial PSD averaged over the frequency annulus assigned to wavelength λ . For each specimen, the endpoint spectral gain was $g(\lambda) = \log_{10}[C_f(\lambda)/C_0(\lambda)]$; $g = 0$ denotes no change and $g = 1$ denotes a tenfold increase in power. The shortest retained annulus required the lower 95% bootstrap bound on g to exceed zero independently in interrupted and uninterrupted experiments, giving $\lambda_{\min} = 4.126 \mu\text{m}$. The upper bound required agreement within 0.10 decade when the same specimens were levelled on the 2×4 and 4×8 fields, bootstrap intervals containing zero in both protocols, and support below the pooled 95th-percentile ACF first-zero crossing. The largest passing annulus was $\lambda_{\max} = 60.871 \mu\text{m}$; the next annulus, $77.131 \mu\text{m}$, failed the section-scale criterion. Beyond this cutoff, the inferred spectral change is not a unique property of the surface: it changes systematically when the same height map is levelled over a different field size, so specimen-scale form and the removed plane cannot be separated from roughness. Extending the comparison to those wavelengths would therefore add operator-dependent whole-field shape rather than experimentally validated roughness information, and a larger numerical surface would not make those unsupported wavelengths suitable for validation.

The large ribbons in Fig. 3 are retained deliberately: they show between-specimen heterogeneity in spectral gain, whereas cutoff selection uses paired within-specimen changes and requires agreement across the two test protocols. Their width therefore limits precision but does not inflate

the number of independent observations. Antialias-filtered resampling of native 50 $\times$ maps showed that 1 μm is the coarsest tested grid providing four cells across $\lambda_{\min}$ without material spectral bias. A periodic lateral domain must contain at least two copies of $\lambda_{\max}$, giving $L \geq 121.74 \mu\text{m}$ and the selected $128 \times 128 \mu\text{m}^2$ surface.

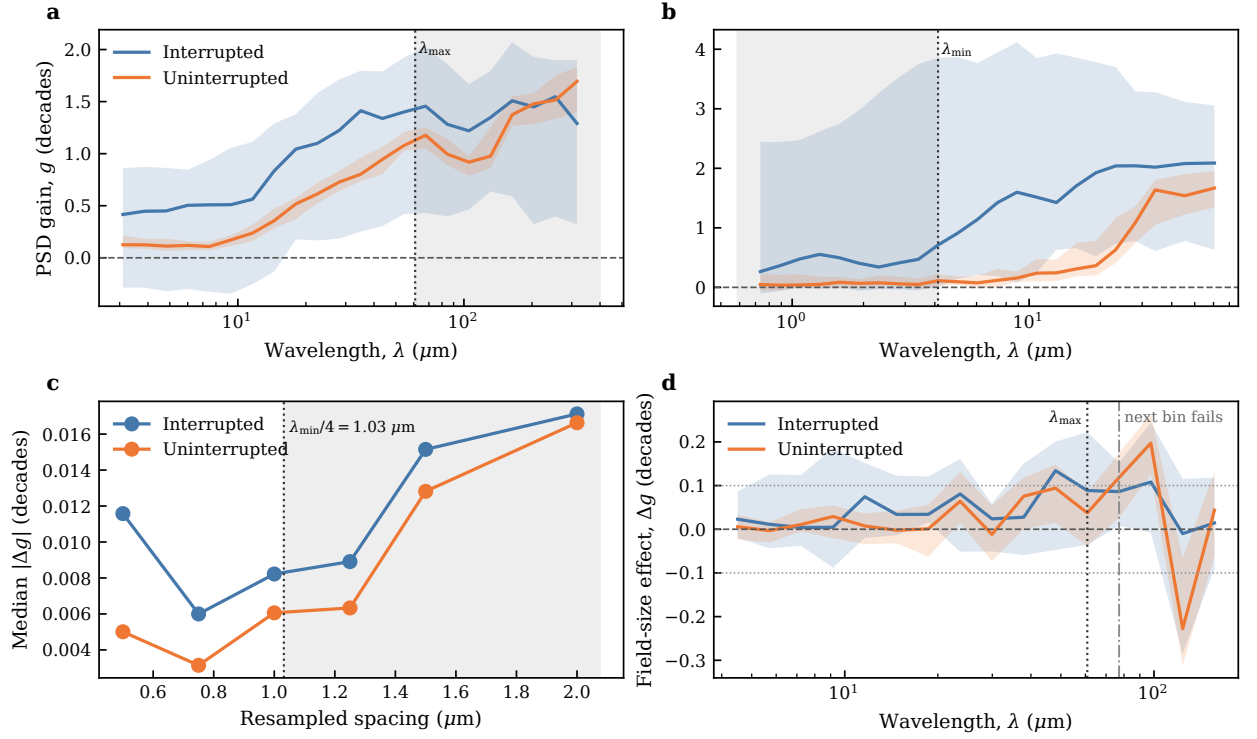


Figure 3: Experimental definition of the retained roughness band. (a,b) Median specimen-paired PSD gain g and 95% bootstrap intervals for 10 $\times$ and 50 $\times$ observations; the dashed line denotes no spectral change. (c) Median absolute error in g after antialias-filtered resampling of native 50 $\times$ maps, shown separately for interrupted and uninterrupted tests. (d) Change in g when the same maps are levelled on 4 $\times$ 8 rather than 2 $\times$ 4 fields. The dotted horizontal limits are ± 0.10 decade; vertical markers identify the retained upper limit and the first longer annulus that fails the field-size test. Together the panels give $4.13 \leq \lambda \leq 60.87 \mu\text{m}$ and a 1 μm production spacing.

2.4. EBSD-informed statistical reconstruction

Orthogonal XY and YZ EBSD maps covered $256.2 \times 204.05 \mu\text{m}^2$ at 0.35 μm spacing and were transformed to a common specimen frame, with z the build direction (Fig. 4). The XZ section was excluded because scratches and pitting obscured a defensible orientation-based segmentation. Plane-specific IQ and misorientation thresholds were used because image quality and surface damage differed between XY and YZ . Pits and scratches were restored to feature ID zero after segmentation and did not enter the grain statistics. A temporary two-pixel closing allowed crystallographically continuous material across the dominant YZ scratch to remain one grain.

Only complete, non-surface grains were merged: 329 from XY and 84 from YZ . The orientation distributions were normalized within each plane and averaged with equal plane weights; misorientations were pooled by measured boundary length. Two-dimensional diameters

and neighbor counts were not treated as direct three-dimensional statistics. Candidate ellipsoidal populations were instead sectioned in both planes and compared with the EBSD diameter, aspect-ratio, neighbor, and major-axis distributions. The accepted population has $\mu_{\ln d} = 2.0$, $\sigma_{\ln d} = 0.85$, $b/a = 0.40$, $c/a = 0.15$, and a $55 \mu\text{m}$ maximum equivalent diameter.

Periodic DREAM.3D realizations were produced in two passes. The first sampled grain geometry; measured XY and YZ major-axis projections then replaced the sampled shape orientations; the fixed geometry was rasterized; and crystallographic orientations were assigned separately from the merged ODF and MDF. This separation prevents grain-shape orientation from replacing the measured texture. The production ensemble contains three independent $128 \times 128 \times 256 \mu\text{m}^3$ realizations with $1 \mu\text{m}$ voxels and an eight-voxel minimum connected feature. The detailed segmentation, forward-sectioning, and realization parameters are listed in Appendix B.

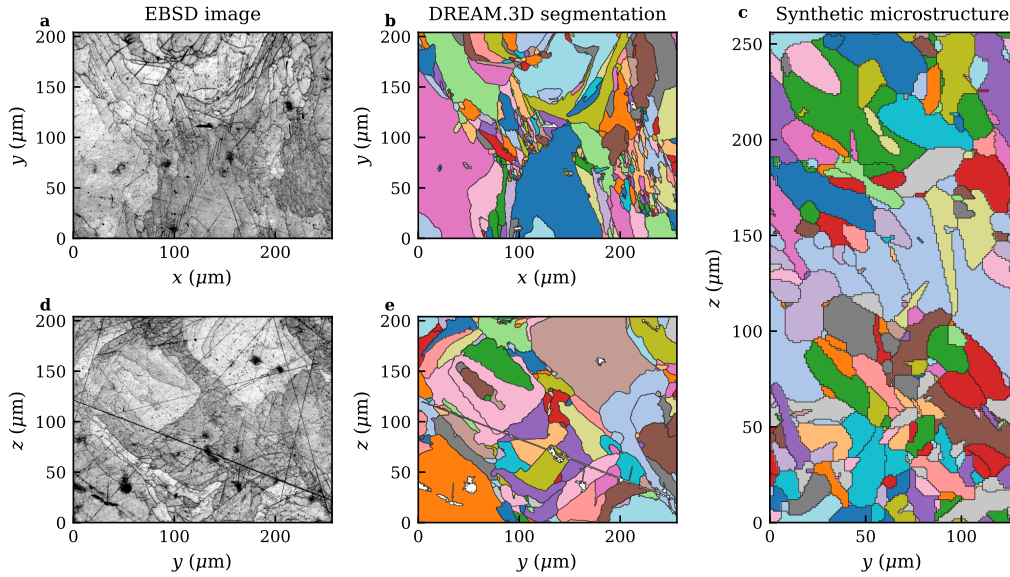


Figure 4: EBSD-informed reconstruction. The first two columns show the measured image-quality fields and DREAM.3D segmentations in the specimen frame for XY (top) and YZ (bottom). White pixels have feature ID zero and do not enter the statistics. The final column is a section through a periodic synthetic realization generated from the merged statistics; z is the build direction.

2.5. Large-strain FFT model and calibration

Room-temperature deformation was simulated with the large-strain elasto-viscoplastic fast Fourier transform (LS-EVPFFT) framework developed in Refs. [15, 16]. The solver advances a periodic voxelized polycrystal while enforcing compatibility through an FFT correction. Plastic flow was represented by FCC crystallographic slip,

$$\mathbf{L}^p = \sum_{\alpha=1}^{N_s} \dot{\gamma}^\alpha \mathbf{s}^\alpha \otimes \mathbf{m}^\alpha, \quad (5)$$

where $\mathbf{s}^\alpha$ and $\mathbf{m}^\alpha$ are the slip direction and slip-plane normal. Slip rates followed

$$\dot{\gamma}^\alpha = \dot{\gamma}_{0,x}^{(\sigma)} \left| \frac{\tau^\alpha}{g^\alpha} \right|^{n_{\text{sr},x}^{(\sigma)}} \text{sign}(\tau^\alpha), \quad (6)$$

where τ^α and g^α are the resolved shear stress and slip resistance. The load-specific reference rate $\dot{\gamma}_{0,x}^{(\sigma)}$ and rate exponent $n_{\text{sr},x}^{(\sigma)}$ parameterize the measured stress range; they are not interpreted as a continuous constitutive law in applied stress. Voce hardening evolved with accumulated slip Γ ,

$$g^\alpha(\Gamma) = \tau_0^\alpha + (\tau_1 + \theta_1 \Gamma) \left[1 - \exp\left(-\frac{\theta_0 \Gamma}{\tau_1}\right) \right], \quad \Gamma = \sum_\alpha \int |\dot{\gamma}^\alpha| dt. \quad (7)$$

The slip resistance combines two distinct strengthening contributions. Voce hardening raises resistance as slip accumulates and therefore represents work hardening during loading and the hold. Hall–Petch strengthening augments the initial resistance by $h_{\text{HP}}/\sqrt{d_g}$, where d_g is the grain diameter, to represent grain boundaries impeding slip transfer. Including both terms prevents the fitted work-hardening law from absorbing a systematic grain-size contribution.

Independent constitutive calibrations were performed for interrupted and uninterrupted protocols. For each candidate and load, the bulk strain histories from three deterministic $32 \times 32 \times 64$ realizations were averaged before calculating the load loss; load losses were then averaged. These calibration realizations occupy the same $128 \times 128 \times 256 \mu\text{m}^3$ physical domain as production but use $4 \mu\text{m}$ voxels. The final orientation-corrected fields contain 1573 grains on average in the calibration ensemble and 1527 in the production ensemble (Table 2). Both exceed the order- 10^3 -grain guideline reported by Lim et al. for stabilizing homogenized uniaxial polycrystal response [17]. This comparison supports the bulk-strain calibration domain; agreement of local morphology is established separately by forward sectioning and the retained experimental roughness bandwidth. Cubic elastic constants were fixed to $C_{11} = 206$, $C_{12} = 133$, and $C_{44} = 119$ GPa for the uninterrupted calibration, whereas the interrupted calibration retained its fitted elastic constants listed in Appendix C. Surface roughness was not part of the objective.

For load ℓ , the simulated target was the mean $\bar{\epsilon}_\ell^{\text{sim}}(t) = \frac{1}{3} \sum_{r=1}^3 \epsilon_{\ell,r}^{\text{sim}}(t)$ of the three microstructures. Simulation and experiment were evaluated on the experimental time grid. The load loss combined an amplitude term and a normalized-shape term,

$$L_\ell = L_{\text{amp},\ell} + L_{\text{shape},\ell}, \quad (8)$$

$$L_{\text{amp},\ell} = \frac{1}{N_\ell} \sum_i \left[\frac{\hat{\epsilon}_\ell(t_i) - \bar{\epsilon}_\ell^{\text{sim}}(t_i)}{\max_t \hat{\epsilon}_\ell(t)} \right]^2, \quad (9)$$

$$L_{\text{shape},\ell} = \frac{1}{N_\ell} \sum_i \left[\frac{\bar{\epsilon}_\ell^{\text{sim}}(t_i)}{\bar{\epsilon}_\ell^{\text{sim}}(t_f)} - \frac{\hat{\epsilon}_\ell(t_i)}{\hat{\epsilon}_\ell(t_f)} \right]^2. \quad (10)$$

The objective was the unweighted mean of L_ℓ over the three loads. Thus stochastic microstructural response was averaged before scoring, in the same order used to construct the experimental population target. Timed-out, failed, or incomplete candidates received the upper loss cap of 2; partial curves were not scored.

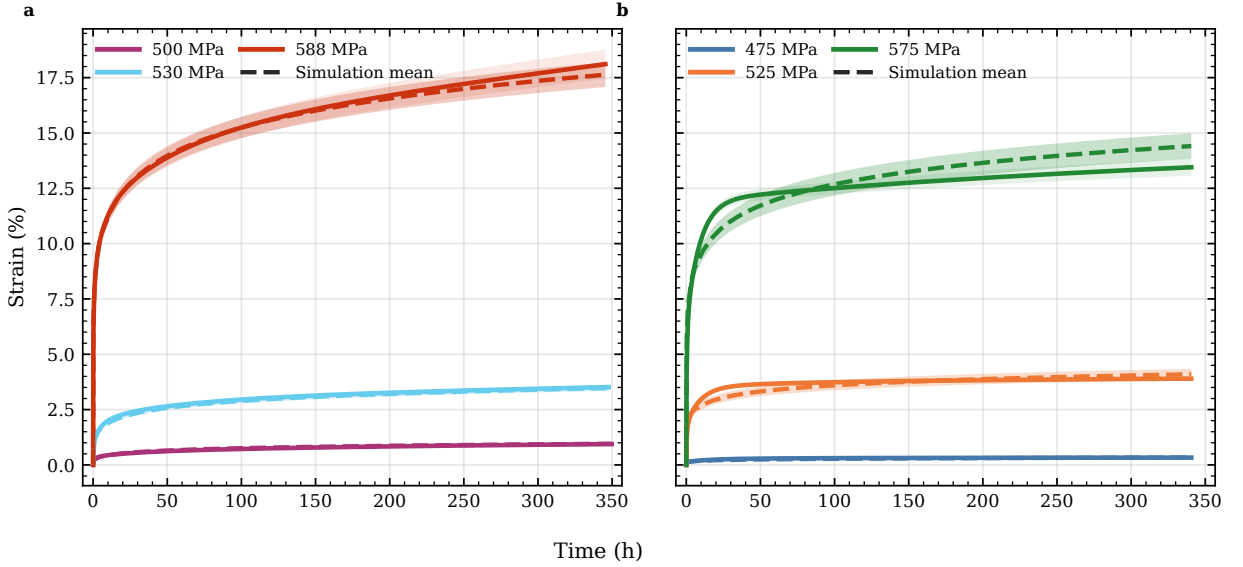


Figure 5: Constitutive calibration fits for (a) uninterrupted calibration 2 and (b) interrupted calibration 3. Solid curves and light bands are the reconstructed experimental targets and their bootstrap standard deviations; dashed curves and darker bands are the mean and standard deviation of the three calibration microstructures. The objective is evaluated from each load's simulated ensemble mean.

The production morphology grid contains $128 \times 128 \times 256$ cells at $1 \mu\text{m}$ spacing. Because the Hall–Petch term scales as $h_{\text{HP}}/\sqrt{d}$, the production coefficient $h_{\text{HP}}^{\text{prod}}$ was set to $2h_{\text{HP}}^{\text{cal}}$ when transferring from the $4 \mu\text{m}$ calibration representation to the $1 \mu\text{m}$ production representation. Four spatial states were retained per load. Normal displacement was extracted from the transverse periodic boundary plane and independently plane levelled. This plane is not traction free; its displacement is therefore a solid-boundary morphology diagnostic rather than an absolute prediction of measured surface height. Endpoint numerical spectra and ACFs are the means of the two unique orthogonal longitudinal boundary planes; opposite periodic faces are not treated as independent observations. For the illustrative 575 MPa comparison, the experimental window was selected by its joint proximity to the cohort medians of endpoint ΔS_a , normalized PSD shape, and ACF shape. Experimental height and numerical displacement were each divided by their own endpoint RMS before applying one common dimensionless color scale. Accumulated slip was summarized by its surface median and 10th–90th percentiles; early-to-final persistence was quantified by Spearman correlation and top-decile retention. The constitutive loss, solver acceptance rules, and matched spatial operators are specified in Appendix C.

2.6. Statistical inference

Interrupted ΔS_a –strain slopes were fitted through the origin after specimen-level baseline subtraction, using every nonbaseline time. Confidence intervals resampled whole specimen histories. Endpoint means, PSD changes, and ACF changes were likewise bootstrapped over specimens; local sections and windows were never treated as independent replicates. Paired spectral changes were tested with two-sided Wilcoxon signed-rank tests and Holm adjustment. Estimator robustness was evaluated under eight prespecified combinations of field size, detrending, taper, bin count, and shortest retained wavelength. The resolved 525 and 575 MPa histories were also fitted to

$\Delta S_a = K\epsilon_c^\beta$. Exponent intervals resampled complete profilometry histories while retaining the population calibration target as the common creep-state coordinate.

3. Results

3.1. Roughness-creep relationship

Figure 6 contains 18 specimens and 126 paired surface-time observations. Fits through the origin after specimen-level baseline subtraction give the results in Table 1. The 525 and 575 MPa specimen-bootstrap intervals exclude zero, whereas the 475 MPa interval crosses zero. Applying the section-level sensitivity filter while retaining median aggregation changes the slopes to 0.102, 0.345, and 0.228 $\mu\text{m}/\%$, respectively, without changing this inference. Across all six candidate section grids, the median endpoint change remained positive for the 525 and 575 MPa interrupted cohorts and the 500 and 530 MPa uninterrupted cohorts; only the unresolved 475 MPa response changed sign. The decrease from the 525 to 575 MPa slope shows that a single stress-independent conversion between roughness and strain is not supported.

Table 1

Specimen-level coupling in the polished interrupted experiments. Final strain is the reconstructed population target; slope intervals resample complete specimen histories.

Condition	n	Final strain (%)	Final ΔS_a (μm)	Slope [95% CI] ($\mu\text{m}/\%$)
475 MPa	6	0.330	0.025	0.101 [−0.067, 0.214]
525 MPa	6	3.898	1.305	0.346 [0.311, 0.385]
575 MPa	6	13.449	3.175	0.257 [0.233, 0.286]

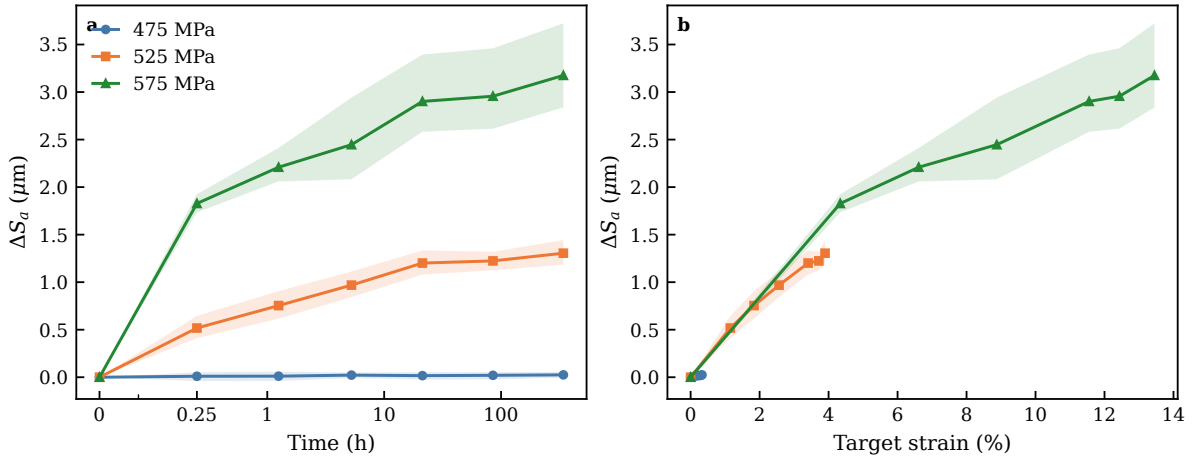


Figure 6: Relationship between surface roughness accumulation and creep deformation in polished interrupted tests. (a) Change in arithmetic mean roughness, $\Delta S_a = S_a(t) - S_a(0)$, versus time. (b) The same change versus the reconstructed population strain target. Curves show specimen-mean roughness and 95% whole-specimen bootstrap intervals. Each surface value is the median of eight independently plane-levelled regions; no section rejection is used in the primary result.

The resolved histories are not proportional relationships between roughness and strain. Fits of

$$\Delta S_a = K\epsilon_c^\beta \quad (11)$$

give $\beta = 0.748$ (95% specimen-bootstrap interval 0.634–0.882, log-space $R^2 = 0.997$) at 525 MPa and $\beta = 0.477$ (0.396–0.586, $R^2 = 0.989$) at 575 MPa. Both intervals lie below the proportional value $\beta = 1$, and the 575 MPa interval lies entirely below the 525 MPa point estimate. By approximately one-third of endpoint strain, the 525 and 575 MPa surfaces have already accumulated 39% and 58% of their endpoint roughness change, respectively (Fig. 7). The surface response is therefore front-loaded: subsequent creep amplifies relief more slowly than the bulk strain accumulates. The 475 MPa mean history is not included because its endpoint ΔS_a is not resolved from zero.

The numerical 525 MPa boundary also develops roughness early relative to its strain, while the uninterrupted 530 MPa experiment supplies only its initial and final surface states. Figure 7 therefore compares the resolved 525 MPa trajectory with the numerical trajectory and shows the 530 MPa endpoint without implying an unmeasured intermediate path.

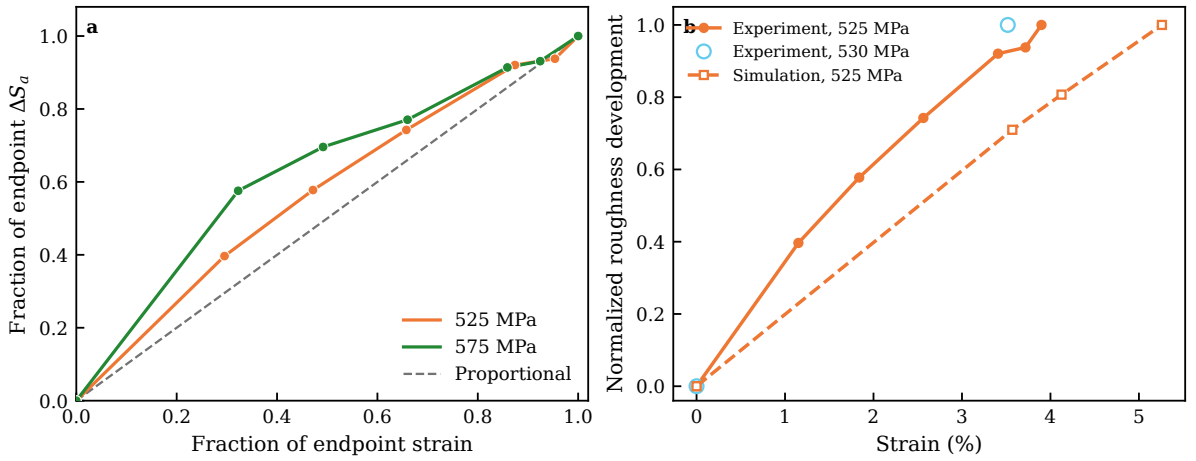


Figure 7: Roughness development relative to strain. (a) Mean interrupted roughness and reconstructed strain normalized by their respective endpoints for the resolved 525 and 575 MPa histories. The dashed diagonal denotes proportional accumulation. (b) Normalized 525 MPa experimental and numerical roughness versus strain, together with the two available 530 MPa uninterrupted states. The comparison shows that roughness develops disproportionately early in both the experiment and the solid-boundary calculation.

The complete stress response is shown in Fig. 8. A common three-parameter monotone curve was fitted to the load-group means,

$$y(\sigma) = A \left\{ \exp \left[B \max(\sigma - \sigma_c, 0) \right] - 1 \right\}, \quad (12)$$

where σ_c is an empirical onset within the tested stress interval, not a measured material threshold. The strain fit spans all six interrupted and uninterrupted conditions and has an RMSE of 0.336 strain-percentage points ($R^2 = 0.997$). Its leave-one-load-out RMSE is 0.56 points, showing that the ordered increase is not produced by a single terminal observation.

The roughness fit uses the five conditions with acquisition-valid paired baselines. Its RMSE is $0.157 \mu\text{m}$ ($R^2 = 0.979$), but the leave-one-load-out RMSE increases to $0.45 \mu\text{m}$. The largest signed residuals occur for the neighboring 525 and 530 MPa conditions: the interrupted value is $0.264 \mu\text{m}$ above the curve and the uninterrupted value is $0.228 \mu\text{m}$ below it. Thus endpoint roughness has

a strong monotone stress dependence, but the protocol offset and sparse load set do not support treating the fitted curve as a universal constitutive relation.

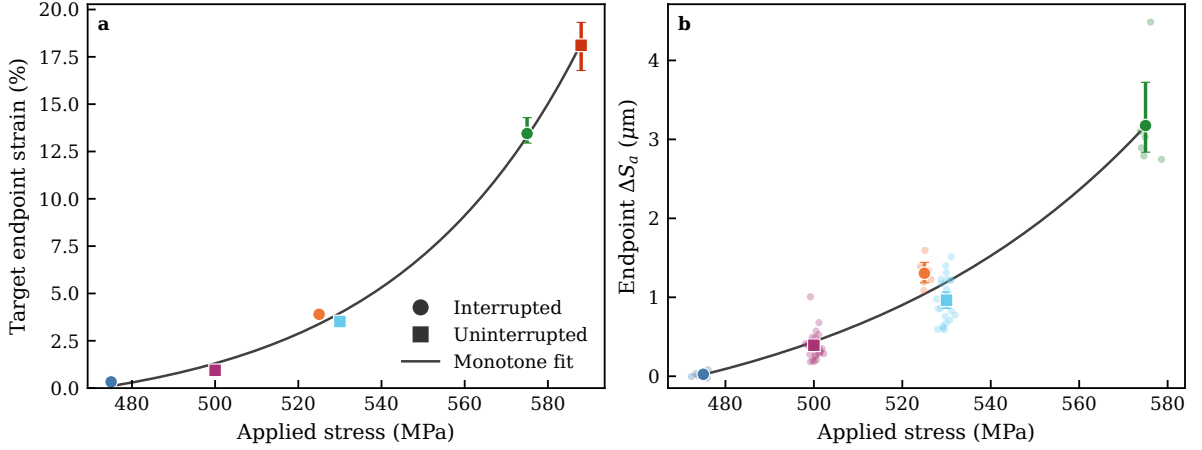


Figure 8: Endpoint response to applied stress. (a) Reconstructed strain target and whole-specimen 95% bootstrap interval for all six conditions. (b) Specimen endpoint roughness change, load-group mean, and 95% specimen-bootstrap interval. The monotone fit uses the five conditions with comparable paired baselines; the 588 MPa condition is omitted because its baseline surface condition differs from the other uninterrupted scans. Lines summarize the observed ordering and are not constitutive laws.

Increasing uninterrupted stress from 500 to 530 MPa raises target endpoint strain from 0.940% (95% interval 0.861–1.029%) to 3.518% (3.402–3.636%) and mean ΔS_a from 0.393 μm (0.330–0.467 μm) to 0.963 μm (0.863–1.069 μm). The corresponding factors of 3.7 and 2.5 provide a handling-free load-level check on the interrupted result. Because the available metadata do not translate DIC ordinals to physical profilometry IDs, this comparison is at the load-group level rather than between registered specimens.

3.2. Spatial redistribution and observation sensitivity

The uninterrupted maps show that creep changes spatial organization as well as scalar amplitude. At 500 MPa, specimen-median RMS height increases from 0.220 to 0.582 μm , spectral median wavelength from 45.1 to 71.0 μm , the fraction of resolved power at $\lambda \geq 32$ μm from 0.592 to 0.823, and radial ACF $1/e$ length from 13.82 to 20.24 μm . The paired median wavelength change is +26.3 μm (95% bootstrap interval 16.66–33.27 μm ; Holm-adjusted $p = 2.45 \times 10^{-5}$), and the ACF-length change is +6.44 μm (5.70–7.48 μm ; $p = 2.47 \times 10^{-5}$).

The 530 MPa evolution is stronger. RMS height increases from 0.519 to 1.390 μm , spectral median wavelength from 20.3 to 57.2 μm , long-wavelength power fraction from 0.324 to 0.700, and ACF length from 4.99 to 14.82 μm . The paired wavelength change is +34.5 μm (28.30–37.00 μm ; Holm-adjusted $p = 7.15 \times 10^{-7}$), and the ACF change is +8.82 μm (8.18–9.76 μm ; $p = 7.15 \times 10^{-7}$). At both loads, power grows preferentially at low frequency and the high-frequency exponent becomes more negative by approximately 0.45. Fine-scale relief is therefore not removed; its contribution grows more slowly than the broad features that increasingly dominate the surface. The field-inconsistent 588 MPa baselines do not enter the primary PSD/ACF evolution analysis.

Surface roughness build-up as an indicator of room-temperature creep

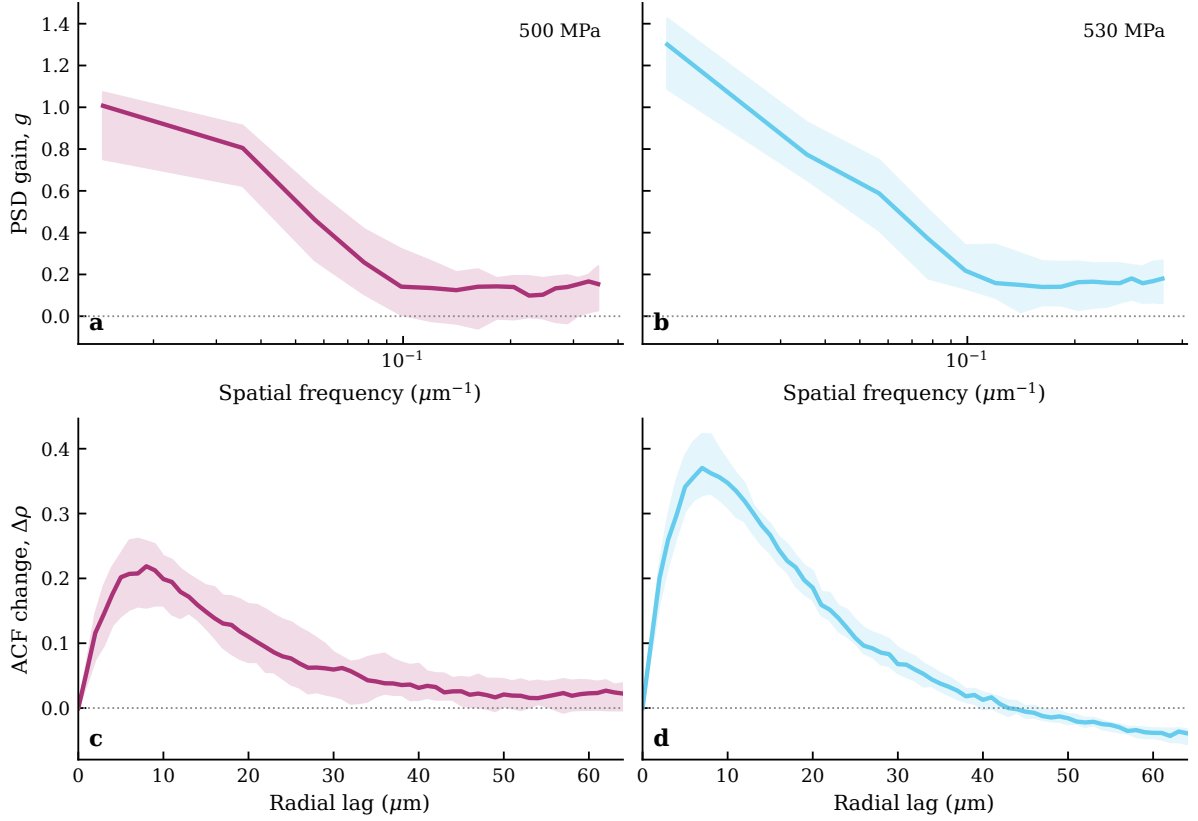


Figure 9: Paired spectral and correlation evolution at 500 and 530 MPa. (a,b) PSD gain, $g(f) = \log_{10}[P_f(f)/P_0(f)]$. (c,d) Change in normalized radial autocorrelation, $\Delta\rho(r) = \rho_f(r) - \rho_0(r)$. Curves are specimen-paired medians with interquartile bands.

The directional broadband result is robust to the declared observation choices. Across all five paired cohorts and eight settings, long–short PSD–gain contrast and long–wavelength power–fraction change remain positive, ACF length change remains positive, and the high–frequency exponent change remains negative. Spectral median wavelength is less stable: its sign is retained under 7/8 settings for interrupted 525 MPa and 8/8 elsewhere. Broadband redistribution, rather than a single median wavelength, is therefore the more defensible spatial descriptor. The interrupted 475 MPa cohort shows the same directional spatial changes even though its scalar roughness–strain slope remains unresolved.

The normalized radial ACF, $\rho(r)$, begins at unity and measures the similarity of heights separated by radial distance r . Its $1/e$ crossing is the first distance at which ρ falls to $e^{-1} = 0.368$ and is reported as a characteristic correlation–decay length. It is not a PSD wavelength and does not set the short–wavelength cutoff. The later first–zero crossing marks the end of positive height correlation and provides the independent upper–scale support check used in Section 2.3.

The endpoint comparison in Fig. 10 separates correlation length from spectral allocation. Experimental ACF $1/e$ lengths span 14.75–20.21 μm , whereas the simulations span the narrower 20.15–21.06 μm range. Five of six numerical values lie within 3.90 μm of experiment; the largest difference is 5.39 μm at uninterrupted 530 MPa. The ACF therefore captures the magnitude of the dominant lateral correlation scale. The normalized PSD does not show equivalent agreement.

Within the retained 4.13–60.87 μm band, experiment–model log-PSD RMSE ranges from 0.71 to 1.43 decades, with the model systematically depleted at short wavelength. Moreover, the mean pairwise inter-load log-PSD separation is 0.329 and 0.316 decades in the interrupted and uninterrupted experiments but only 0.061 decades in either simulation set. The periodic solid calculation consequently produces an almost load-invariant normalized morphology: it recovers an overall correlation length while missing both fine-scale power and much of the experimental load-to-load spectral diversity.

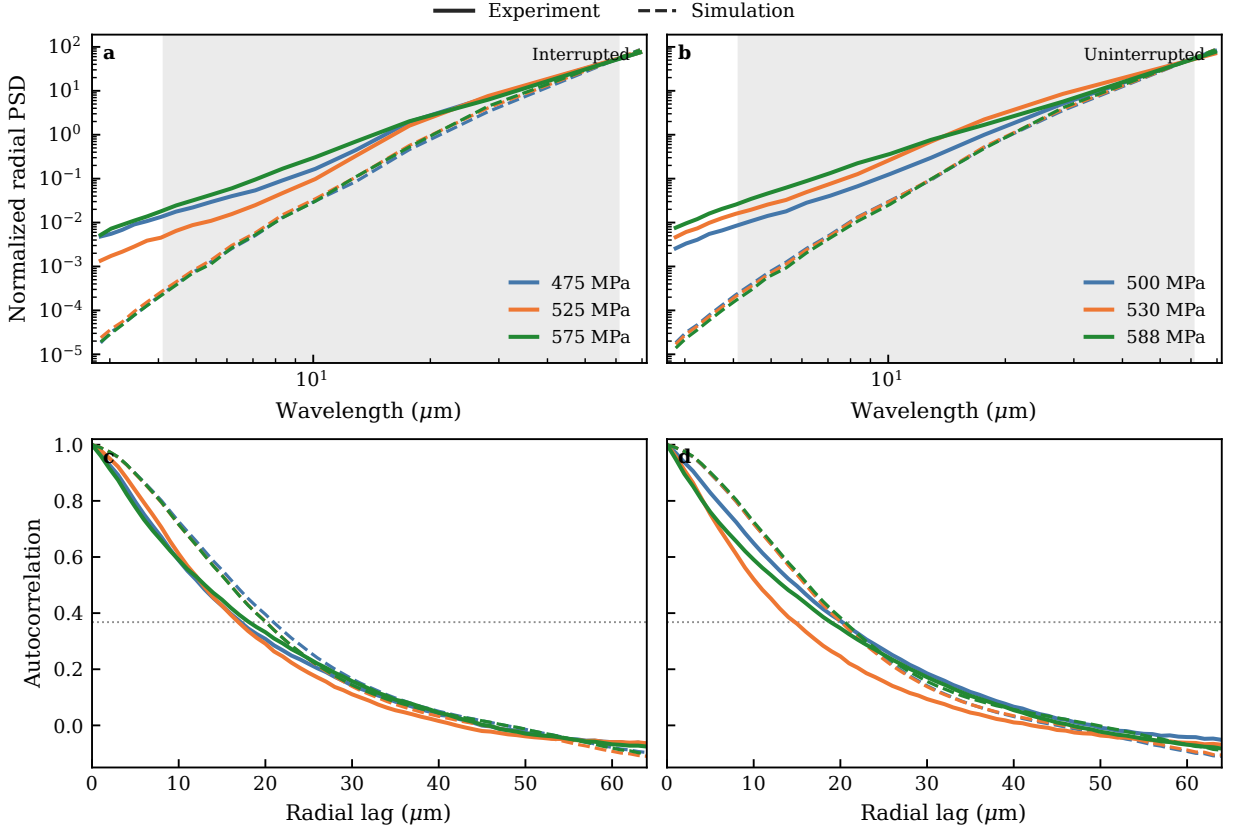


Figure 10: Endpoint spatial organization in all six experiments and corresponding solid-boundary simulations. (a,b) Radial PSD normalized to unit resolved power. The gray region is the experimentally retained 4.13–60.87 μm wavelength band. (c,d) Radial ACF; the horizontal dotted line marks $1/e$. Experimental curves are solid and numerical curves are dashed. Numerical curves average the two unique orthogonal longitudinal boundary planes; opposite periodic faces are duplicates and are not counted separately. The 588 MPa curve is its acquisition-valid post-test endpoint and is not a pre/post evolution result.

The paired objectives separate a robust evolution result from operator-dependent absolute spectra. In fiducial-excluded peripheral patches at 530 MPa, the post/pre long-minus-short PSD-gain contrast is positive in 21/25 specimens at 10 $\times$ and 25/25 at 50 $\times$. The median contrasts are 0.296 decades (0.223–0.492; Holm-adjusted $p = 4.17 \times 10^{-6}$) and 0.906 decades (0.702–1.118; $p = 1.79 \times 10^{-7}$), respectively. The corresponding post-test fractions of resolved 2.76–32 μm

power are 0.749 at 10 $\times$ and 0.477 at 50 $\times$. Thus both objectives reproduce preferential broad-feature growth, but the absolute spectral allocation is objective dependent.

3.3. Representative morphology and slip evolution

The 575 MPa interrupted condition provides the clearest repeated-field comparison without using the unreliable 588 MPa baseline. Specimen 32c, window 31, minimizes the mean percentile distance to the cohort medians in endpoint ΔS_a , normalized PSD shape, and ACF shape; it was not chosen by visual inspection. Figure 11 compares that fixed experimental window with the corresponding numerical boundary footprint. The experimental endpoint RMS height is 3.376 μm and the numerical endpoint RMS displacement is 0.482 μm . Each row is divided by its own endpoint RMS, so the shared color scale displays development and morphology without asserting absolute-amplitude agreement.

Most experimental contrast appears by the first 5.25-h interruption, after which broad positive and negative regions remain the dominant visual scale. The numerical boundary likewise establishes its broad pattern early and primarily increases its contrast thereafter. Because the microscope field was repositioned at each interruption, individual ridges are not tracked between experimental panels; the sequence instead shows the population-scale development of broad relief.

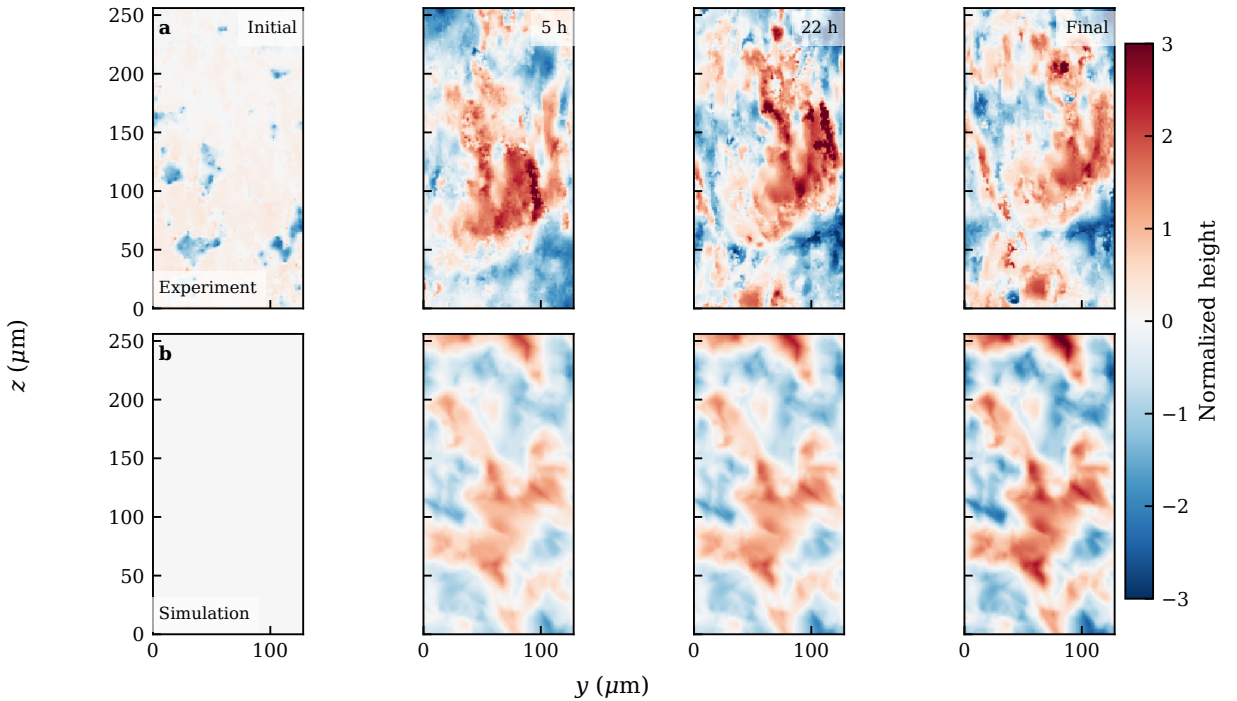


Figure 11: Representative 575 MPa height-field evolution. (a) Plane-levelled experimental height in specimen 32c, window 31. (b) Plane-levelled normal displacement on the numerical x boundary. The experimental and numerical endpoint RMS amplitudes are 3.376 and 0.482 μm , respectively, so each row is divided by its endpoint RMS to compare morphology development on a common normalized scale.

The accumulated-slip fields explain why the numerical morphology changes mainly by amplification (Fig. 12). Between 5.20 and 353.28 h, the slip-field rank correlation is 0.978 and 87.3% of cells in the early top decile remain in the final top decile. The localization index, $\langle \Gamma^2 \rangle / \langle \Gamma \rangle^2$,

decreases only from 1.052 to 1.042 while median accumulated slip increases from 0.255 to 0.404. Consequently, the late hold does not create a new hotspot population; it amplifies a spatial template selected during the early transient and weakly broadens the participating area.

Accumulated slip co-localizes with equivalent plastic strain ($\rho = 0.74$ – 0.78 after the initial state), but not with the magnitude of normal boundary displacement ($\rho = 0.03$ – 0.04). Roughness is therefore not a pointwise surrogate for total slip. The boundary relief instead reflects the compatibility of heterogeneous slip among neighboring grains, which is consistent with the persistent broad displacement pattern and the failure of the solid model to reproduce the experimental short-wavelength PSD.

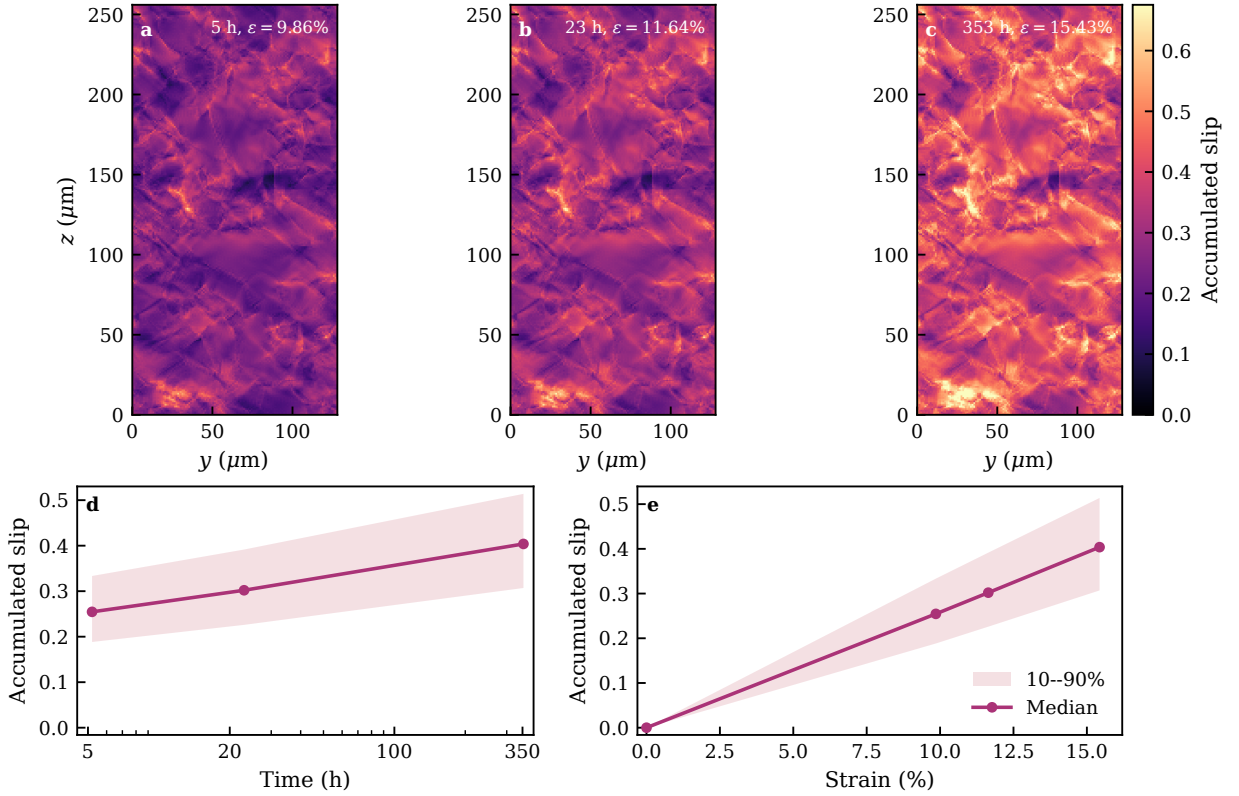


Figure 12: Accumulated slip on the paired 575 MPa numerical x boundary. (a–c) Spatial fields at the three nonzero retained states; one absolute color scale is used throughout. (d) Surface median and 10th–90th percentiles versus time. (e) The same distribution versus simulated strain. Spatial ranks and early hotspots persist while the magnitude increases.

4. Discussion

4.1. What surface roughness indicates

The interrupted and uninterrupted datasets answer different parts of the indicator question. Repeated interrupted measurements resolve longitudinal amplitude but may include handling, unloading, repositioning, and reloading effects. Uninterrupted tests remove those cycles and independently reproduce the load ordering of strain and roughness growth. Their paired surface maps also show that creep shifts power preferentially toward broad features. Agreement between

these designs supports a baseline-to-follow-up indicator without implying that a single terminal roughness value determines strain across loads.

Baseline information is essential because the 500 and 530 MPa cohorts begin with markedly different spectra. Creep changes both amplitude and lateral organization: RMS height rises, long-wavelength power grows preferentially, and ACF length increases. These directions survive all eight estimator settings and both microscope objectives, whereas the magnitude of spectral allocation does not. A practical indicator should therefore combine an amplitude change such as ΔS_a with a baseline-referenced broadband redistribution measure. Spectral median wavelength alone is less defensible because its sign is estimator sensitive in one cohort.

The sublinear exponents add a temporal constraint to that indicator. A fixed increment of roughness does not represent a fixed increment of strain: roughening efficiency is largest during the early transient and declines as the hold proceeds. The decline is stronger at 575 MPa, even though its absolute roughness change is larger. This behavior is consistent with early selection of heterogeneous deformation paths followed by continued slip on that established network. It also explains why endpoint amplitude alone cannot distinguish histories that reach similar S_a by different stress paths. Repeated measurements supply information not contained in a single terminal map.

The stress dependence of the experimental slopes and the degradation from specimen-held-out to load-held-out prediction prevent a universal conversion from roughness to strain. Additional independent builds, geometries, surface preparations, registered DIC/profilometry identifiers, and load histories are required before roughness can be used as a quantitative creep-state estimator. The present result is an association within the tested material and stress window, not a lifetime model.

4.2. Model scope and limitations

The numerical observation has a narrower interpretation than the experimental surface measurement. The periodic solid boundary is not traction free, so its normal displacement cannot be compared to profilometry as an absolute height field. Plane levelling, baseline subtraction, normalization, and identical spatial operators permit tests of relative amplitude, wavelength redistribution, and field persistence, but agreement in any one of these quantities does not establish a free-surface prediction. The 4 μm calibration ensemble and 1 μm production representation also require the grain-size strengthening term to be expressed on a consistent physical length scale, as specified in the Methods.

The numerical comparison identifies a more specific shortfall than an amplitude mismatch. Its ACF decay is close to the measured endpoint correlation scale, and its slip and displacement patterns are highly persistent. However, its normalized PSD is too concentrated at long wavelength and changes little among loads. The missing short-wavelength power cannot be repaired by multiplying displacement by a scalar. It points instead to physics or boundary conditions that create fine relief in the experiment but are absent from a periodic solid plane, including a traction-free surface, surface-grain relaxation, and experimentally present subgrain-scale heterogeneity. The near-zero pointwise correlation between accumulated slip and displacement magnitude further shows that a local slip-amplitude rule is insufficient; compatibility and slip gradients must enter any free-surface extension.

Experimental limitations include a single coupon geometry and build condition, no spatial registration of pre/post profilometry, incomplete uninterrupted cross-modality identifiers, no independent acquisition-axis metadata, and incomplete and field-inconsistent pre-test profilometry at 588 MPa. Large between-specimen bands are retained because they quantify real population heterogeneity and observed missingness; local windows are never promoted to independent specimens to make those bands artificially smaller. The 1 μm , eight-cell production reconstruction resolves the EBSD section statistics and the retained experimental roughness band. The two unregistered EBSD sections do not identify the joint three-dimensional distribution of grain shape and orientation, however. These limitations preclude a unique local slip mechanism or a transferable roughness-based life prediction.

5. Conclusions

Surface roughness contains a reproducible but non-proportional record of room-temperature creep in LPBF 316L stainless steel. Interrupted 525 and 575 MPa tests resolve nonzero longitudinal roughness–strain slopes, whereas the 475 MPa response remains within low-load scatter. Exponents of 0.748 and 0.477 show that roughness develops disproportionately early and becomes more front-loaded at higher stress. Uninterrupted 500 and 530 MPa tests independently confirm increasing roughness, spectral wavelength, long-wavelength power fraction, and ACF length. Baseline-referenced amplitude, broadband redistribution, and measurement history are therefore complementary indicators, but the available data do not define a stress-independent or externally validated roughness-to-strain calibration.

The EBSD-informed reconstruction reproduces the measured median section diameters and aspect ratios while preserving the specimen coordinate frame. Forward sectioning is essential because the two-dimensional diameter, shape, and neighbor distributions are not direct three-dimensional statistics. The remaining plane-specific shape differences and the absence of registered three-dimensional EBSD constrain subsequent local-field interpretation.

The experimentally retained 4.13–60.87 μm band defines a $128 \times 128 \mu\text{m}^2$ lateral numerical domain and 1 μm sampling without equating microscope resolution with a physical roughness cutoff. Forward-sectioned EBSD statistics and three deterministic calibration realizations provide the microstructural and mechanical baseline for spatial comparison. The solid-boundary simulations reproduce the experimental correlation-length scale and show that slip hotspots are selected early and persist, but their PSDs contain too little short-wavelength power and too little load-to-load diversity. A traction-free surface treatment, registered uninterrupted measurements, additional builds and surface preparations, and three-dimensional microstructural characterization remain necessary before translating these associations into a transferable creep-state or lifetime predictor.

A. Experimental procedure and data processing

A.1. Specimen, loading, and measurement geometry

The Renishaw AM400 build used a 50 μm layer thickness, 110 μm hatch spacing, 200 W interior laser power, 0.75 m s^{-1} scan speed, and a meander pattern. Two border passes used 110 W and 0.2 m s^{-1} . The multi-gauge specimen dimensions, serial loading fixture, and build parameters are summarized in Fig. 1. DIC images were acquired with 12.3-MP monochrome FLIR Grasshopper 3 cameras and processed with VIC-2D. The first image after loading was the DIC

reference, so reported strain excludes elastic loading and inelastic strain accumulated during the loading ramp. Profilometry used a Keyence VK-X3000; native lateral spacings were $1.380 \mu\text{m}$ at $10\times$ and $0.278 \mu\text{m}$ at $50\times$, with 768×1024 height samples per export.

A.2. DIC target reconstruction

Let $\epsilon_{s,i}$ be the measured strain of specimen s at time i and S_i the specimens finite at both ends of interval i . The composition-stable population increment is

$$\Delta\bar{\epsilon}_i = \frac{1}{|S_i|} \sum_{s \in S_i} (\epsilon_{s,i} - \epsilon_{s,i-1}), \quad (13)$$

and the target history is accumulated from these increments. Cross-reset increments following 0.25, 1.25, 5.25, 21.25, and 85.25 h are zeroed in the interrupted data. The smooth target is

$$\hat{\epsilon}(t) = \sum_{k=1}^{24} a_k [1 - \exp(-t/\tau_k)] + bt, \quad a_k \geq 0, \quad b \geq 0, \quad (14)$$

where the fixed τ_k span 20 s to three experiment durations. The fit is rescaled to the paired-increment endpoint. The target is continuous, monotone, and has a non-increasing transient rate plus a linear asymptote; it does not insert terminal acceleration to replace an unobserved specimen tail.

The endpoints are 0.330, 3.898, and 13.449% for the interrupted 475, 525, and 575 MPa targets and 0.940, 3.518, and 18.111% for the uninterrupted 500, 530, and 588 MPa targets. Their whole-specimen bootstrap 95% intervals are 0.271–0.434, 3.714–4.111, 12.946–14.287, 0.861–1.029, 3.402–3.636, and 16.779–19.319%, respectively. Figure 2 separates changes caused by cohort composition and DIC resets from the final smoothing step.

A.3. Height-map validity, sectioning, and levelling

Every CSV was required to have the recorded XY calibration, exactly 768×1024 height values, a missing fraction no greater than 10^{-4} , and no connected missing region larger than one pixel. Isolated one-pixel export gaps were nearest-neighbor filled. All 444 maps passed; 31 contained one filled pixel.

The native $10\times$ map was divided into a 2×4 grid, and each 384×256 -pixel ($530 \times 353 \mu\text{m}^2$) section was independently plane levelled. The map statistic was the median section S_a . This operator was the only tested partition to provide at least eight subsamples, a short side spanning ten times the conservative 95th-percentile ACF $1/e$ length ($31.55 \mu\text{m}$), and at least 50 circular correlation areas. It provides 59.9 correlation areas and 11.2 correlation lengths on the short side. The median relative bootstrap standard error is 15.7%.

A modified-z section screen based on both S_a and maximum absolute height was evaluated but not used. It removed 226 of 1808 sections and preferentially selected high-relief material: rejected and retained median S_a values were 2.010 and $0.528 \mu\text{m}$. Four maps lost all sections. With all acquisition-valid sections retained, the leave-one-section-out change of the map median is 2.8% at the median and 13.6% at the 95th percentile. Figure 13 reports the complete selection and sensitivity result.

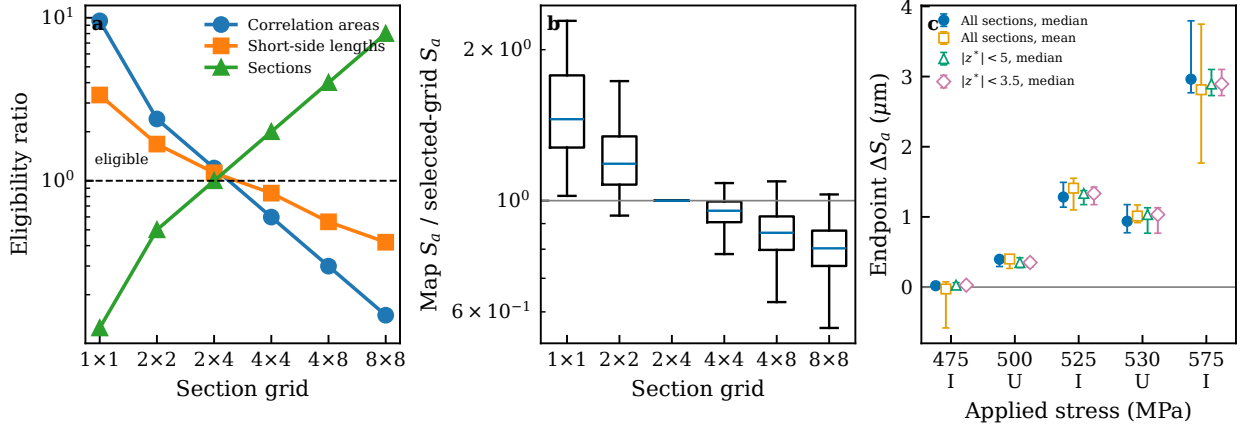


Figure 13: Sectioning and response-screen sensitivity. (a) Physical eligibility ratios for each partition: correlation area, short-side length, and number of sections are divided by their required minima. The dashed line at $10^0 = 1$ is the pass threshold, and only the 2×4 partition satisfies all three criteria. (b) Distribution of map S_a relative to the selected-grid value; ratios below one show the progressive removal of long-scale form by finer levelling. (c) Endpoint changes under median and mean aggregation and under the two modified-z screens. The primary result is the median of all acquisition-valid sections.

A.4. Retained roughness bandwidth and numerical domain

The lower cutoff is not the microscope resolution. For each specimen and frequency annulus, the paired evolution measure was

$$g(\lambda) = \log_{10} [C_f(\lambda)/C_0(\lambda)] . \quad (15)$$

The shortest retained annulus required the lower 95% bootstrap bound on g to exceed zero independently in the interrupted and uninterrupted groups. This first occurs at $\lambda_{\min} = 4.126 \mu\text{m}$; both intervals include zero at the next shorter annulus, $3.406 \mu\text{m}$.

The upper cutoff was determined by repeating the paired PSD calculation with the 2×4 and 4×8 levelling operators on the same specimens. A candidate required, in each protocol, an absolute median fine-minus-primary change no greater than 0.10 decades and a 95% interval containing zero. It also could not exceed the pooled 95th percentile of the endpoint ACF first-zero crossing, $77.103 \mu\text{m}$. The largest passing annulus is $\lambda_{\max} = 60.871 \mu\text{m}$. At this wavelength the interrupted and uninterrupted median section effects are 0.089 and 0.037 decades, with intervals $[-0.036, 0.222]$ and $[-0.038, 0.084]$. The wide interrupted interval limits precision but does not manufacture acceptance: the next annulus, $77.131 \mu\text{m}$, has a consistently signed section effect and fails.

Native 50× fields were antialias-filtered and resampled to candidate spacings. A $1 \mu\text{m}$ grid is the coarsest tested grid providing four cells across $\lambda_{\min}$; its median absolute final/initial spectral-gain error is 0.007 decades and final-PSD error is 0.013 decades. A periodic lateral domain must contain two copies of $\lambda_{\max}$ so the upper wavelength is not represented only by the fundamental mode. Thus $L \geq 2\lambda_{\max} = 121.74 \mu\text{m}$, rounded upward to $128 \mu\text{m}$. The retained band, section-scale check, ACF evidence, and resolution test are shown in Figs. 3 and 14.

The 588 MPa pre-test maps contain directional grinding relief, do not resolve the intended fiducial, and have a median S_a 10.5 and 4.9 times the 500 and 530 MPa baselines. Only 14 specimen

Surface roughness build-up as an indicator of room-temperature creep

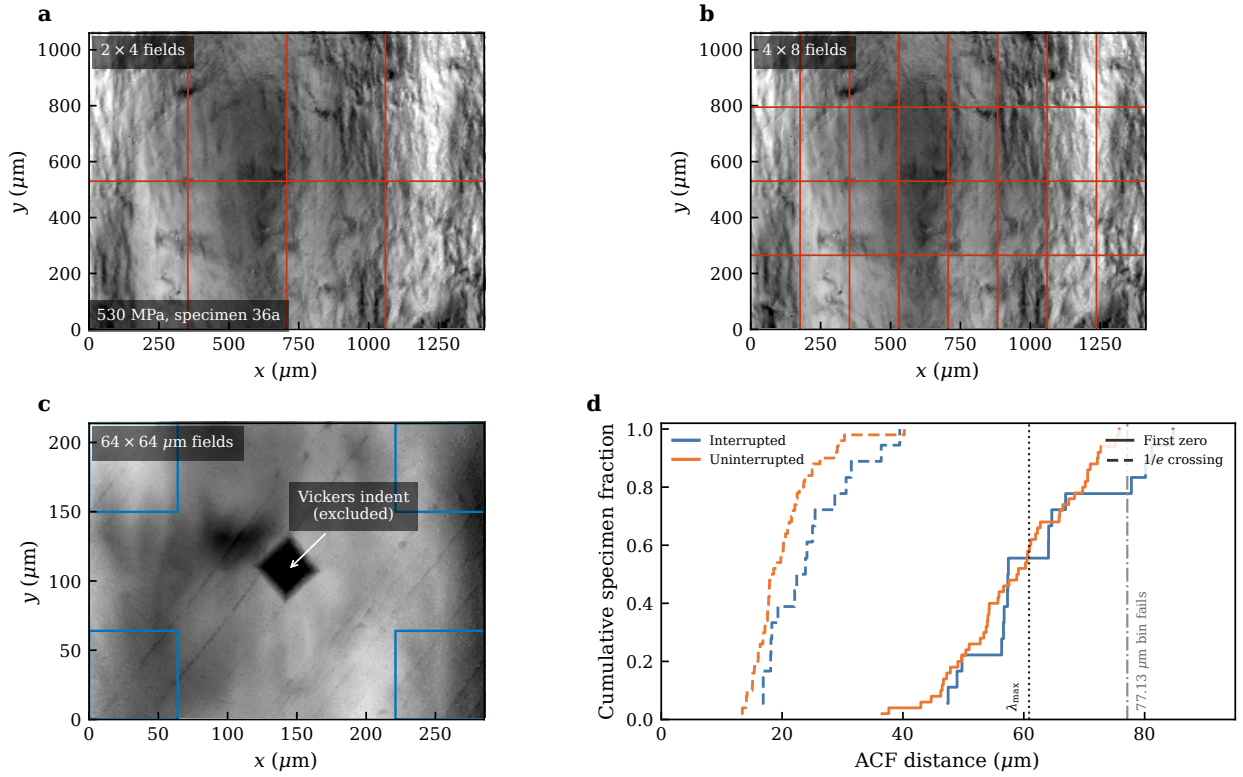


Figure 14: Supporting evidence for the retained 4.13–60.87 μm band. (a,b) The same 10 $\times$ height map partitioned into the selected 2 $\times$ 4 fields and the finer 4 $\times$ 8 fields used for the levelling-scale sensitivity test. (c) Four 64 $\times$ 64 μm^2 50 $\times$ fields used for the resolution test; the central Vickers indent is excluded. (d) Empirical cumulative distributions of specimen ACF first-zero and 1/ e distances for the two protocols. The 1/ e distance measures correlation decay, whereas the first zero bounds the extent of positive height correlation. The retained $\lambda_{\max} = 60.871 \mu\text{m}$ lies below the pooled 95th-percentile first-zero distance and passes the paired section-scale test; the next annulus at 77.131 μm fails that test.

identifiers form pre/post pairs. These maps are retained in the common directory layout and in scalar sensitivity outputs, but they do not enter cross-load or paired PSD/ACF inference.

B. EBSD treatment and synthetic microstructure

The XY and YZ maps cover $256.2 \times 204.05 \mu\text{m}^2$ at $0.35 \mu\text{m}$ spacing. CI thresholds are 0.10 in both planes and IQ thresholds are 14000 and 12000. Bad-data and neighbor-correlation misorientation tolerances are 3° for XY and 5° for YZ ; grain-growth tolerances are 1.5° and 6° . Both planes use a 16-pixel minimum segmented feature and two-neighbor minimum. A temporary two-pixel in-plane closing bridges the dominant YZ scratch during segmentation; the measured defect mask is restored afterward. Cleanup uses only the two physical image directions. Feature ID zero, inactive phases, and edge-intersecting grains do not enter the merged morphology statistics.

Production realizations are periodic in all directions and use a corrected two-pass procedure: DREAM.3D first samples geometry; measured XY and YZ major-axis projections then replace the sampled shape orientations in a realization-specific feature description; the fixed realization is rasterized; and crystallographic orientations are assigned separately from the merged ODF and MDF. Figure 4 shows the measured IQ fields, accepted segmentations, and a section through the merged-statistics microstructure.

The three final calibration fields contain 1638, 1546, and 1534 grains; the three production fields contain 1508, 1520, and 1553 grains. Counts are unique positive feature IDs in the final orientation-corrected DREAM.3D fields after the second rasterization, rather than counts from the provisional sampled fields. Table 2 summarizes the two ensembles. Lim et al. found that approximately 1000 or more grains were required to reduce the realization-to-realization standard deviation of uniaxial stress to about 1 MPa at 10% strain in morphology-bearing polycrystals, and concluded that an order of 10^3 grains is a practical uniaxial-response RVE guideline [17]. The present counts exceed that bulk-response benchmark. The citation does not establish convergence of local surface fields; that requirement is instead addressed by the forward-section match and the experimental wavelength limits.

Table 2

Realized grain counts in the final orientation-corrected microstructures. Each ensemble contains three independent realizations.

Ensemble	Grid	Spacing (μm)	Mean grains [range]
Calibration	$32 \times 32 \times 64$	4	1573 [1534–1638]
Production	$128 \times 128 \times 256$	1	1527 [1508–1553]

C. Calibration and numerical analysis

Two material parameterizations were retained: calibration 2 for the uninterrupted 500, 530, and 588 MPa experiments and calibration 3 for the interrupted 475, 525, and 575 MPa experiments. Calibration 2 used fixed cubic elastic constants; calibration 3 included the cubic constants in the fitted parameter set. Tables 3 and 4 list the values used in the final calculations.

The calibration Hall–Petch value is tied to grain diameters expressed on the $4 \mu\text{m}$ calibration grid. Production evaluates the same grain-size term on $1 \mu\text{m}$ grid. To preserve the Hall–Petch shift, $h_{\text{HP}}/\sqrt{d_g}$, the production coefficient was set to $h_{\text{HP}}^{\text{prod}} = 2h_{\text{HP}}^{\text{cal}}$. All other parameters transfer without grid scaling.

For calibration 2, both m and $\log_{10} \dot{\gamma}_{0,x}$ increase nearly linearly with applied stress ($R^2 = 1.000$ and 0.998 ; Fig. 15). The connecting segments summarize the three independently calibrated values

Table 3

Load-independent material parameters retained from calibrations 2 and 3. The two Hall–Petch rows distinguish the coefficient fitted on the calibration grid from the coefficient transferred to the production grid.

Parameter	Units	Calibration 2, uninterrupted	Calibration 3, interrupted
τ_{0f}	MPa	141.593	154.337
τ_{0b}	MPa	126.056	146.376
τ_1	MPa	2.858	14.711
θ_0	MPa	2990.660	1005.130
θ_1	MPa	891.677	398.338
C_{11}	GPa	206.000	197.506
C_{12}	GPa	133.000	150.224
C_{44}	GPa	119.000	114.655
h_{HP}^{cal}	MPa $\mu m^{1/2}$	165.645	81.462
h_{HP}^{prod}	MPa $\mu m^{1/2}$	331.290	162.925
h_{self}	–	1.000	1.000
h_{latent}	–	1.400	1.400

Table 4

Load-specific reference shear rate and rate sensitivity. The rate-sensitivity exponent is $n_{sr,x}$ and $m = 1/n_{sr,x}$.

Calibration	Stress (MPa)	$\dot{\gamma}_{0,x}$ (s^{-1})	$n_{sr,x}$	m
2, uninterrupted	500	4.188×10^{-6}	40.577	0.025
2, uninterrupted	530	3.161×10^{-5}	19.589	0.051
2, uninterrupted	588	3.888×10^{-3}	9.492	0.105
3, interrupted	475	4.917×10^{-8}	79.129	0.013
3, interrupted	525	3.596×10^{-4}	42.908	0.023
3, interrupted	575	1.450×10^{-3}	18.376	0.054

and do not prescribe intermediate-load behavior. Calibration 3 values are retained in Table 4 but omitted from the figure so the uninterrupted trend remains legible.

The calibration objective and ensemble-averaging order are defined in Section 2.5. A recorded loss is capped above at 2.000. Timed-out, failed, or incomplete candidates receive the cap; partial curves are not scored. The asynchronous search keeps eight candidates in flight, with nine six-thread simulations per three-load candidate.

For a plane-levelled field h , the Hann-tapered periodogram is

$$P(k_x, k_z) = \Delta x \Delta z \frac{|\mathcal{F}\{w(h - \bar{h})\}|^2}{\sum w^2}. \quad (16)$$

Radial bins are physical-frequency annuli. The ACF uses zero-padded linear convolution and divides covariance by the number of overlapping pairs before zero-lag normalization. The numerical

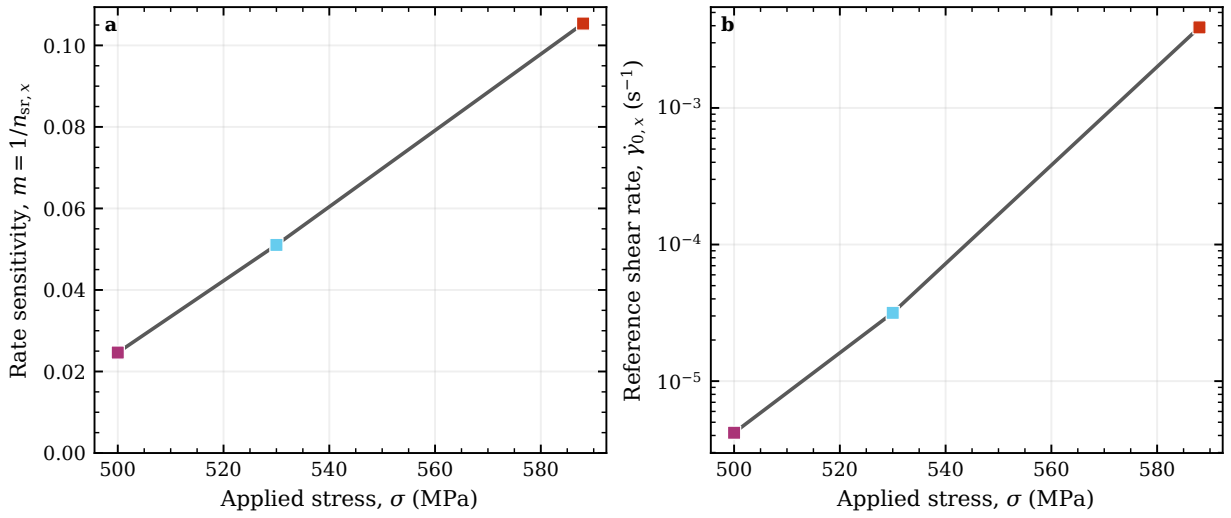


Figure 15: Stress dependence of the uninterrupted calibration-2 rate parameters. (a) Rate sensitivity $m = 1/n_{sr,x}$. (b) Reference shear rate on a logarithmic ordinate. Symbols are independently calibrated load-specific values; lines show their stress trend.

and experimental operators use the same physical frequency limits; upsampling experimental data never extends the native measurement bandwidth.

CRedit authorship contribution statement

Gavin DeBrun: Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Nathan K. Brown:** Conceptualization, Methodology, Software, Data curation, Writing - original draft. **Samuel Inman:** Methodology, Investigation, Data curation, Writing - original draft. **Ricardo Lebensohn:** Methodology, Software, Writing - original draft. **Brad L. Boyce:** Conceptualization, Investigation, Writing - original draft. **Rémi Dingreville:** Conceptualization, Methodology, Writing - original draft.

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